Code No: 158AU **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech IV Year II Semester Examinations, July/August - 2022
DISTRIBUTED SYSTEMS
(Common to CSE, IT)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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1.a) Explain the challenges in distributed systems.
b) Explain the generations of distributed systems. [7+8]

2.a) Explain about communication distributed objects.
b) Explain remote procedure call (RPC). [8+7]

3.a) Explain the core operating system components and their responsibilities.
b) Enumerate the differences between thread and a process. Explain the thread scheduling. [7+8]

4.a) Explain the requirements of a distributed file system.
b) Explain the flat file service operations. [7+8]

5.a) Explain the functional and non-functional requirements of peer-to-peer middleware.
b) Give a short note on message routing infrastructure: PASTRY. [8+7]

6. What is distributed mutual exclusion? Explain any one algorithm to implement mutual exclusion. [15]

7.a) What is a transaction? Explain nested transactions.
b) Explain about the communication in two-phase commit protocol. [7+8]

8.a) Explain the passive (primary-backup) model for fault tolerance.
b) Explain in which respects DSM is suitable or unsuitable for client-server systems. [7+8]

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